Michael Estrada

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SUMMARY

Software engineer with 15+ years of experience designing, building, and operating developer tools, CI/CD systems, and release infrastructure at scale. Technical lead and core contributor to a large-scale event-driven release platform, building Go microservices, Temporal workflow orchestration, and federated GraphQL infrastructure. Experienced in applying AI to developer workflows through MCP-based tooling and intelligent automation. Deep expertise across the full stack, from Kubernetes orchestration to developer-facing CLIs and dashboards.

SKILLS

Languages: Go, Python, TypeScript, JavaScript, Java, Groovy, C#, SQL, Shell
AI / Developer Tooling: Model Context Protocol (MCP), LLM integration (Gemini, Claude, GPT-4), prompt engineering and evaluation, RAG-based knowledge retrieval, AI-assisted code generation and review workflows
Infrastructure & Orchestration: Kubernetes, Helm, Containers, Temporal, Harness, Jenkins, Ansible, Pulumi, AWS EKS
Data & Messaging: GraphQL (Apollo Federation), NATS, Kafka, CloudEvents, MySQL, PostgreSQL, MSSQL
Platforms & Protocols: Linux/UNIX, mTLS/TLS, Git, Artifactory, Vault

PROFESSIONAL EXPERIENCE

Apple Inc., Sunnyvale, CA

Software Engineer — Release Engineering & Developer Infrastructure | July 2018 – Present

Technical lead for a developer experience platform, driving the architecture, adoption, and evolution of services and workflows that power the dev-test-release lifecycle for consumer-facing products across four business units.

Release Infrastructure & Workflow Orchestration

- Led architecture and development of 10+ Go microservices and Temporal workflows powering a content release pipeline serving 9 product categories across 20+ device families, including artifact upload, content distribution, deployment automation, and security scanning.
- Technical lead for a Go-based event routing service that dispatches CloudEvents to Temporal workflow namespaces, enabling multi-tenant release orchestration across teams.
- Led design and implementation of a high-throughput Go service handling artifact ingestion, validation, and distribution across pre-production and production environments, increasing one team's release cadence from 1–2 per week to multiple daily deployments.
- Drove adoption of multi-cluster Kubernetes deployments (AWS EKS), designing release workflows spanning dev, UAT, and production environments using Helm and Harness pipelines.

Federated GraphQL & Apollo Router Platform

- Led the design and implementation of a GitOps-driven Apollo Router platform, automating schema validation, supergraph recomposition, and router deployment across multiple Kubernetes clusters.
- Led development of Go-based workflows for PQL (Persisted Query List) validation and schema composition, integrating with GitHub check runs to provide real-time developer feedback on pull requests.
- Designed and deployed an mTLS sidecar proxy for secure service-to-service authentication across federated subgraph services.
- Drove federated data graph configuration strategy across environments, enabling safe, incremental schema changes without service disruption.

AI-Powered Developer Tools

- Led development of MCP (Model Context Protocol) servers integrating LLM-based AI assistants with project management and documentation systems, enabling autonomous CRUD operations, format conversion, and contextual retrieval.

- Designed and implemented prompt pipelines and RAG-based knowledge retrieval, integrating LLM APIs (Gemini, Claude, GPT-4) into developer workflows for automated code review, test generation, and release documentation.
- Authored domain-specific skills and knowledge bases for an AI agent platform, enabling LLM-driven tools to reason about GitOps configuration patterns, deployment topologies, and release procedures.

CI/CD Pipelines & Build Infrastructure

- Led long-term ownership of core CI/CD pipeline repositories, spanning Jenkins pipeline definitions, build tooling, and deployment automation across CMS, retail, channel, and interactive properties supporting 9 product categories and 20+ device families — including 3D modeling and XR content teams.
- Built and maintained SVN tooling supporting content repository migrations, handling edge cases in large-scale SCM transitions.
- Defined and tracked release health metrics across teams, building a deployment tracking service (TypeScript) and analytics dashboard with API gateway (Node.js) that provided data-driven insights into deployment velocity and pipeline reliability.
- Created CLI tools in Node.js and Python utilized within CI pipelines, streamlining development and release workflows.
- Automated a previously manual data pipeline — replacing scripts and hand-managed uploads with an hourly scheduled workflow — eliminating hours of manual effort per cycle and enabling reliable, repeatable delivery.
- Authored reusable Helm charts and Kubernetes configurations adopted across multiple service deployments.
- Managed infrastructure updates using Ansible playbooks to ensure systems met internal security standards.

Testing Frameworks & Developer Experience (2018–2020)

- Led the development of dynamic testing frameworks for consumer-facing products, leveraging Node.js (Mocha, Selenium, TestCafe) with container-based portability.
- Developed an authentication library bridging test harnesses with internal identity management environments.
- Crafted Jenkins pipelines using Groovy, integrating test harnesses for continuous integration and automated reporting.

Prior Experience

Consultant — Excelon Development (Remote) | 2018 · **Senior Support Engineer** — American Residential Warranty, Boca Raton, FL | 2016–2018 · **QMO Test Engineer** — American Airlines, Phoenix, AZ | 2014–2015 · **Senior QA Associate** — Publicis.Sapient, Miami, FL | 2007–2012

Earlier roles spanning test automation, CI/CD, infrastructure management, and build engineering across enterprise environments. Built testing frameworks (Selenium, C#/.NET, Python), automated server configurations, managed pre-production environments with Ansible, and led build/release processes using Python, PowerShell, and MSBuild.